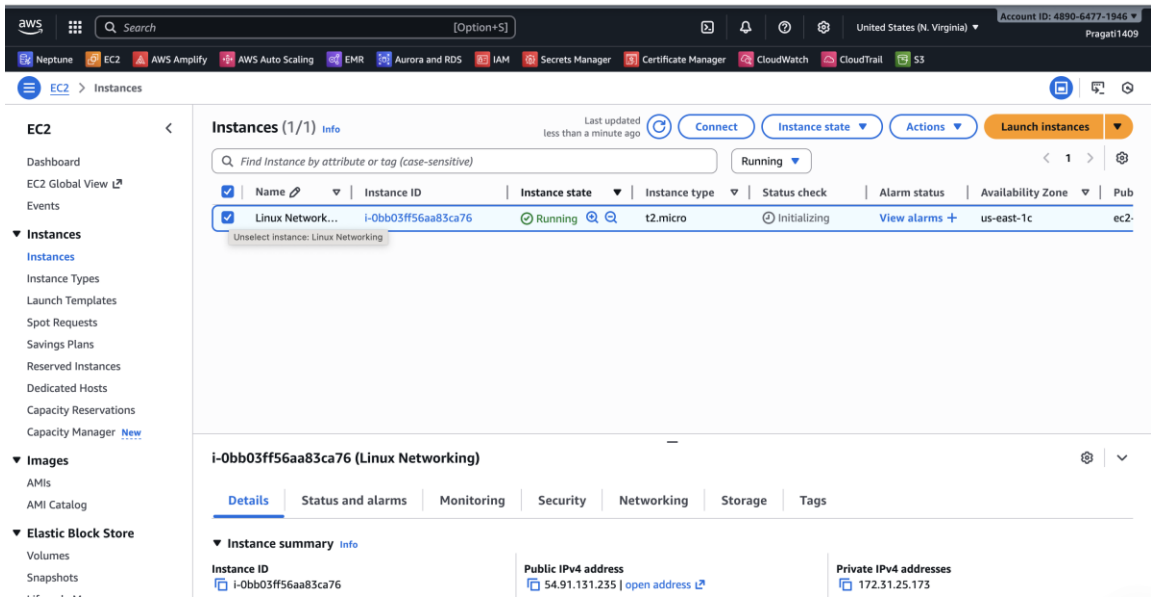
MSML650 – Lab 6

Name: Pragati Rao (UID: 121332776)

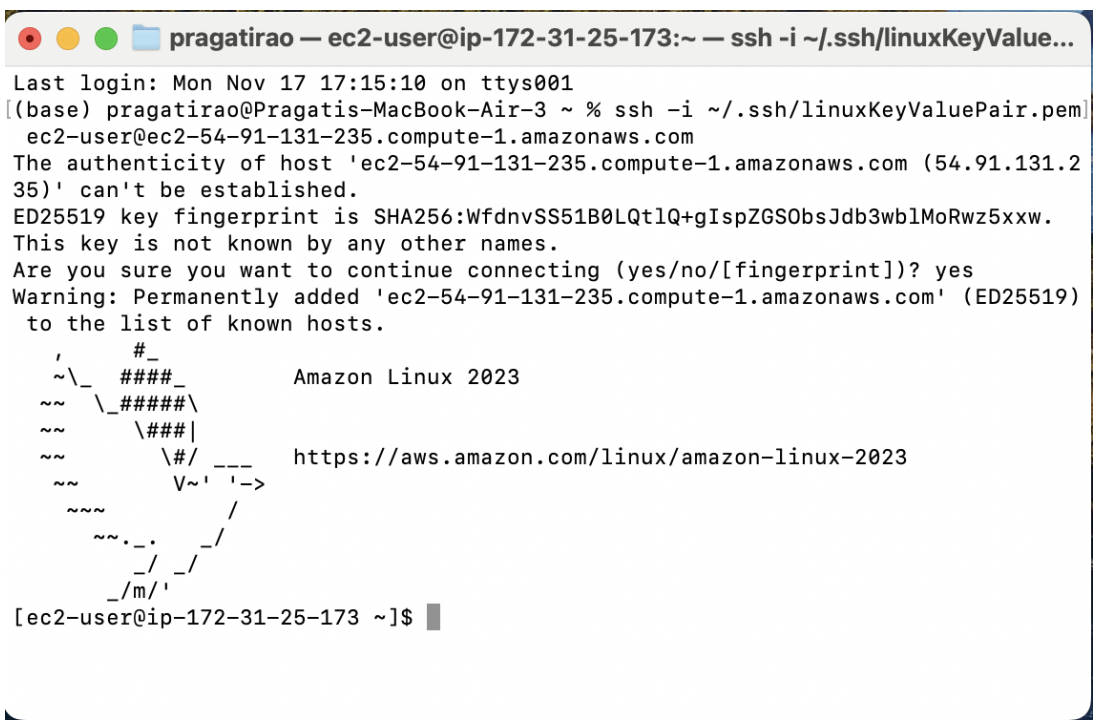
Linux Networking

A. Preparation

1. EC2 instance



2. SSH session



```
pragatirao — ec2-user@ip-172-31-25-173:~ — ssh -i ~/.ssh/linuxKeyValue...
Last login: Mon Nov 17 17:15:10 on ttys001
(base) pragatirao@Pragatis-MacBook-Air-3 ~ % ssh -i ~/.ssh/linuxKeyValuePair.pem
 ec2-user@ec2-54-91-131-235.compute-1.amazonaws.com
The authenticity of host 'ec2-54-91-131-235.compute-1.amazonaws.com (54.91.131.2
35)' can't be established.
ED25519 key fingerprint is SHA256:WfdnvSS51B0LQtlQ+gIspZGSObsJdb3wblMoRwz5xxw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-54-91-131-235.compute-1.amazonaws.com' (ED25519)
 to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-25-173 ~]$
```

B. Linux commands

3. Install the bind-utils package

pragatirao — ec2-user@ip-172-31-25-173:~ — ssh -i ~/.ssh/linuxKeyValuePair.pem...

```
[ec2-user@ip-172-31-25-173 ~]$ sudo yum install -y bind-utils
Amazon Linux 2023 Kernel Livepatch repository   218 kB/s |  29 kB     00:00
Package bind-utils-32:9.18.33-1.amzn2023.0.4.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-25-173 ~]$ host
Usage: host [-aCdilrTvVw] [-c class] [-N ndots] [-t type] [-W time]
            [-R number] [-m flag] [-p port] hostname [server]
       -a is equivalent to -v -t ANY
       -A is like -a but omits RRSIG, NSEC, NSEC3
       -c specifies query class for non-IN data
       -C compares SOA records on authoritative nameservers
       -d is equivalent to -v
       -l lists all hosts in a domain, using AXFR
       -m set memory debugging flag (trace|record|usage)
       -N changes the number of dots allowed before root lookup is done
       -p specifies the port on the server to query
       -r disables recursive processing
       -R specifies number of retries for UDP packets
       -s a SERVFAIL response should stop query
       -t specifies the query type
       -T enables TCP/IP mode
       -U enables UDP mode
       -v enables verbose output
       -V print version number and exit
       -w specifies to wait forever for a reply
       -W specifies how long to wait for a reply
       -4 use IPv4 query transport only
       -6 use IPv6 query transport only
[ec2-user@ip-172-31-25-173 ~]$
```

4. Resolve the "google.com" domain to IPv4 and IPv6 addresses.

pragatirao — ec2-user@ip-172-31-25-173:~ — ssh -i ~/.ssh/linuxKeyValuePair.pem...

```
[ec2-user@ip-172-31-25-173 ~]$ host -t A google.com
google.com has address 192.178.218.101
google.com has address 192.178.218.138
google.com has address 192.178.218.139
google.com has address 192.178.218.100
google.com has address 192.178.218.102
google.com has address 192.178.218.113
[ec2-user@ip-172-31-25-173 ~]$ host -t AAAA google.com
google.com has IPv6 address 2607:f8b0:4004:c21::8a
google.com has IPv6 address 2607:f8b0:4004:c21::66
google.com has IPv6 address 2607:f8b0:4004:c21::64
google.com has IPv6 address 2607:f8b0:4004:c21::71
[ec2-user@ip-172-31-25-173 ~]$
```

5. Issue a DNS query only for the NS records

pragatirao — ec2-user@ip-172-31-25-173:~ — ssh -i ~/.ssh/linuxKeyValuePair.pem...

```
[ec2-user@ip-172-31-25-173 ~]$ host -t NS google.com
google.com name server ns4.google.com.
google.com name server ns3.google.com.
google.com name server ns1.google.com.
google.com name server ns2.google.com.
[ec2-user@ip-172-31-25-173 ~]$
```

6. Check the DNS server's IP address

pragatirao — ec2-user@ip-172-31-25-173:~ — ssh -i ~/.ssh/linuxKeyValuePair.pem...

```
[ec2-user@ip-172-31-25-173 ~]$ host -t NS google.com
google.com name server ns4.google.com.
google.com name server ns3.google.com.
google.com name server ns1.google.com.
google.com name server ns2.google.com.
[ec2-user@ip-172-31-25-173 ~]$ cat /etc/resolv.conf
# This is /run/systemd/resolve/resolv.conf managed by man:systemd-resolved(8).
# Do not edit.
#
# This file might be symlinked as /etc/resolv.conf. If you're looking at
# /etc/resolv.conf and seeing this text, you have followed the symlink.
#
# This is a dynamic resolv.conf file for connecting local clients directly to
# all known uplink DNS servers. This file lists all configured search domains.
#
# Third party programs should typically not access this file directly, but only
# through the symlink at /etc/resolv.conf. To manage man:resolv.conf(5) in a
# different way, replace this symlink by a static file or a different symlink.
#
# See man:systemd-resolved.service(8) for details about the supported modes of
# operation for /etc/resolv.conf.

nameserver 172.31.0.2
search ec2.internal
[ec2-user@ip-172-31-25-173 ~]$
```

7. Display the IP addresses and prefix lengths

pragatirao — ec2-user@ip-172-31-25-173:~ — ssh -i ~/.ssh/linuxKeyValuePair.pem...

```
[ec2-user@ip-172-31-25-173 ~]$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 10
00
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enX0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc fq_codel state UP group default
 qlen 1000
    link/ether 0a:ff:c8:cd:f4:b7 brd ff:ff:ff:ff:ff:ff
    altname eni-04e99391e930880d1
    altname device-number-0.0
    inet 172.31.25.173/20 metric 512 brd 172.31.31.255 scope global dynamic enX0
       valid_lft 2886sec preferred_lft 2886sec
    inet6 fe80::8ff:c8ff:fecd:f4b7/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
[ec2-user@ip-172-31-25-173 ~]$
```

8. Display the link-layer network traffic statistics

```
pragatirao — ec2-user@ip-172-31-25-173:~ — ssh -i ~/.ssh/linuxKeyValuePair.pem...
[ec2-user@ip-172-31-25-173 ~]$ ip -s link show enX0
2: enX0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc fq_codel state UP mode DEFAULT
group default qlen 1000
    link/ether 0a:ff:c8:cd:f4:b7 brd ff:ff:ff:ff:ff:ff
    RX:  bytes packets errors dropped  missed   mcast
      53360548   37423      0       0       0       0
    TX:  bytes packets errors dropped carrier collsns
        290623    3817      0       0       0       0
    altname eni-04e99391e930880d1
    altname device-number-0.0
[ec2-user@ip-172-31-25-173 ~]$
```

Answer: There are **0 errors**, **0 dropped packets**, and the interface state is **UP**.

Number of Packet Received (RX): 37,423 packets

Number of Packet Sent (TX): 3,817 packets

Both RX and TX show **0 errors**, **0 dropped**

9. Show the OS's routing table on your EC2 instance

```
pragatirao — ec2-user@ip-172-31-25-173:~ — ssh -i ~/.ssh/linuxKeyValuePair.pem...
[ec2-user@ip-172-31-25-173 ~]$ ip route
default via 172.31.16.1 dev enX0 proto dhcp src 172.31.25.173 metric 512
172.31.0.2 via 172.31.16.1 dev enX0 proto dhcp src 172.31.25.173 metric 512
172.31.16.0/20 dev enX0 proto kernel scope link src 172.31.25.173 metric 512
172.31.16.1 dev enX0 proto dhcp scope link src 172.31.25.173 metric 512
[ec2-user@ip-172-31-25-173 ~]$
```

10. Verify that the default gateway (router) is reachable

```
pragatirao — ec2-user@ip-172-31-25-173:~ — ssh -i ~/.ssh/linuxKeyValuePair.pem...
[ec2-user@ip-172-31-25-173 ~]$ ping -c 4 172.31.16.1
PING 172.31.16.1 (172.31.16.1) 56(84) bytes of data.
64 bytes from 172.31.16.1: icmp_seq=1 ttl=127 time=0.305 ms
64 bytes from 172.31.16.1: icmp_seq=2 ttl=127 time=0.799 ms
64 bytes from 172.31.16.1: icmp_seq=3 ttl=127 time=0.136 ms
64 bytes from 172.31.16.1: icmp_seq=4 ttl=127 time=0.541 ms

--- 172.31.16.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3100ms
rtt min/avg/max/mdev = 0.136/0.445/0.799/0.249 ms
[ec2-user@ip-172-31-25-173 ~]$
```

11. Given the information obtained above, fill out the following table summarizing your instance's networking configuration:

eth0 network interface's IPv4 address and network prefix length	172.31.25.173/20
eth0 network interface's IPv6 address and network prefix length	fe80::8ff:c8ff:fecd:f4b7/64
Default gateway's (next-hop router's) IPv4 address	172.31.16.1
DNS server's IPv4 address	172.31.0.2

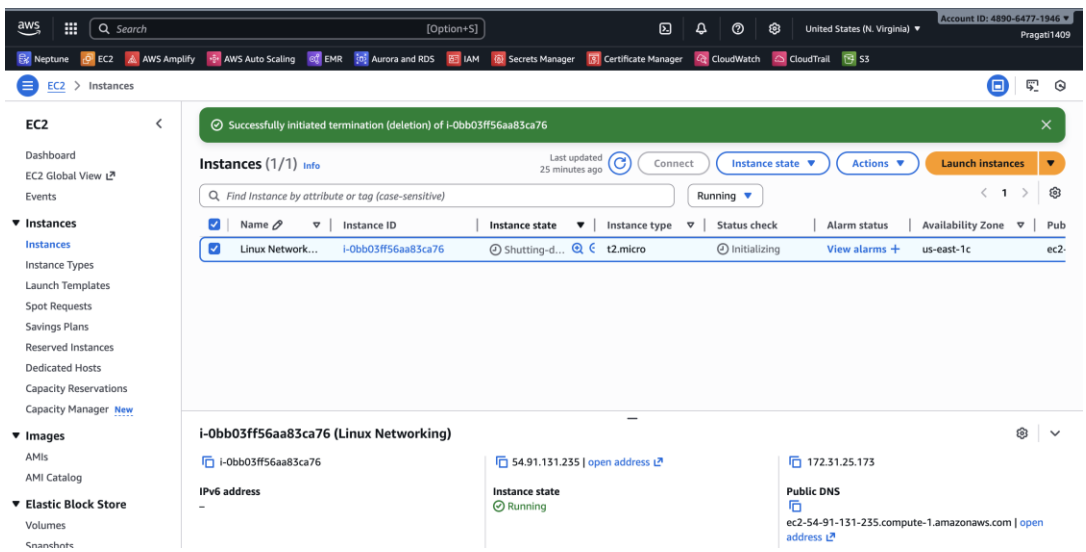
C. Clean up after yourself

12. Terminate the EC2 instance

pragatirao — ec2-user@ip-172-31-25-173:~ — -zsh — 89×31

```
[ec2-user@ip-172-31-25-173 ~]$ ping -c 4 172.31.16.1
PING 172.31.16.1 (172.31.16.1) 56(84) bytes of data.
64 bytes from 172.31.16.1: icmp_seq=1 ttl=127 time=0.305 ms
64 bytes from 172.31.16.1: icmp_seq=2 ttl=127 time=0.799 ms
64 bytes from 172.31.16.1: icmp_seq=3 ttl=127 time=0.136 ms
64 bytes from 172.31.16.1: icmp_seq=4 ttl=127 time=0.541 ms

--- 172.31.16.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3100ms
rtt min/avg/max/mdev = 0.136/0.445/0.799/0.249 ms
[ec2-user@ip-172-31-25-173 ~]$ exit
logout
Connection to ec2-54-91-131-235.compute-1.amazonaws.com closed.
(base) pragatirao@Pragatis-MacBook-Air-3 ~ %
```



13. Log out of AWS